Chapter 4 Truncation Errors and Taylor Series

Taylor Series:

- Provides a mean of predicting a function value at one pt. i.t.o the function value and its derivative at another point.
- Thm states that any smooth function can be approximated as a polynomial.

Taylor's Thm

$$f(x) = f(a) + f'(a)(x-a) + \frac{1}{2!}f''(a)(x-a)^2 + \frac{1}{3!}f'''(a)(x-a)^3 + \cdots$$

($f(x) \to x_{i+1}$, $f(a) \to x_i$)

$$+ \frac{1}{n!}f^{(n)}(a)(x-a)^n + R_n$$

$$R_n = \int_a^x \frac{(x-t)^n}{n!} f^{(n+1)}(t)\,dt \quad , t: \text{dummy variable}$$

Assumptions: $f(x)$ and its first $(n+1)$ derivatives are continuous on an interval containing a and x.

Term by Term

first term $f(x_{i+1}) \cong f(x_i)$ → zero order approximation

second term $f(x_{i+1}) \cong f(x_i) + f'(x_i)(x_{i+1}-x_i)$ → first-order approximation

straight line

$$f(x_{i+1}) \cong f(x_i) + f'(x_i)(x_{i+1}-x_i) + \frac{1}{2!}f''(x_i)(x_{i+1}-x_i)^2 + \cdots$$

h (stepsize)

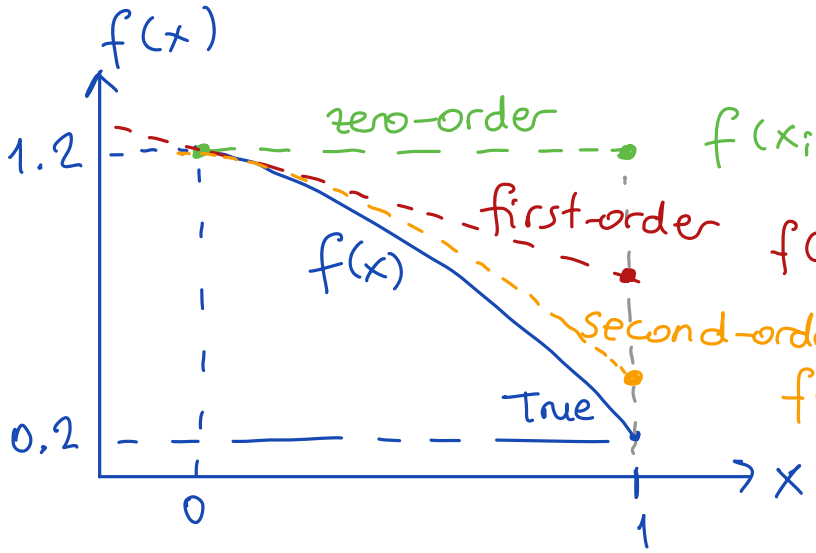
$$+ \frac{f^{(n)}(x_i)}{n!}(x_{i+1}-x_i)^n + R_n$$

$$R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!}\underbrace{(x_{i+1}-x_i)}_{h:\text{stepsize}}^{n+1} = O(h^{n+1})$$, where ξ is a pt. btwn x_i and x_{i+1}

EX $f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$

Use zero to 4th order Taylor series expansion to approximate the function.

$x_i = 0$ with $h = 1 \Rightarrow$ Predict $f(x)$ at $x = 1$



$f(x_{i+1}) \cong f(x_i) \Rightarrow f(1) = f(0)$

$f(x_{i+1}) \cong f(x_i) + f'(x_i)h$

$f(x_{i+1}) \cong f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2$

$f(0) = 1.2$, $f(1) = 0.2$ ↝ True value we're trying to predict

order	Approximation	Truncation Error at $x=1$
$n=0$	$f(1) \cong 1.2$	$E_t = 0.2 - 1.2 = -1$
$n=1$	$f'(x)\Big\|_{x=0} = -(0.4)x^3 - 0.45x^2 - x - 0.25\Big\|_{x=0}$	
	$f'(0) = -0.25$	
	$f(1) \cong 1.2 - 0.25h$	
	$f(1) \cong 0.95$	$E_t = 0.2 - 0.95 = -0.75$
$n=2$	$f''(0) = -1$	
	$f(x_{i+1}) \cong 1.2 - 0.25h - 0.5h^2$	$E_t = 0.2 - 0.45 =$
	$f(1) = 0.45$	-0.25

* $R_n = O(h^{n+1})$ → error is proportional to h^{n+1}

e.g. if error is $O(h)$, halving the step size h, will halve the error.

if error is $O(h^2)$, halving the step size, error will quarter the error

Numerical Differentiation

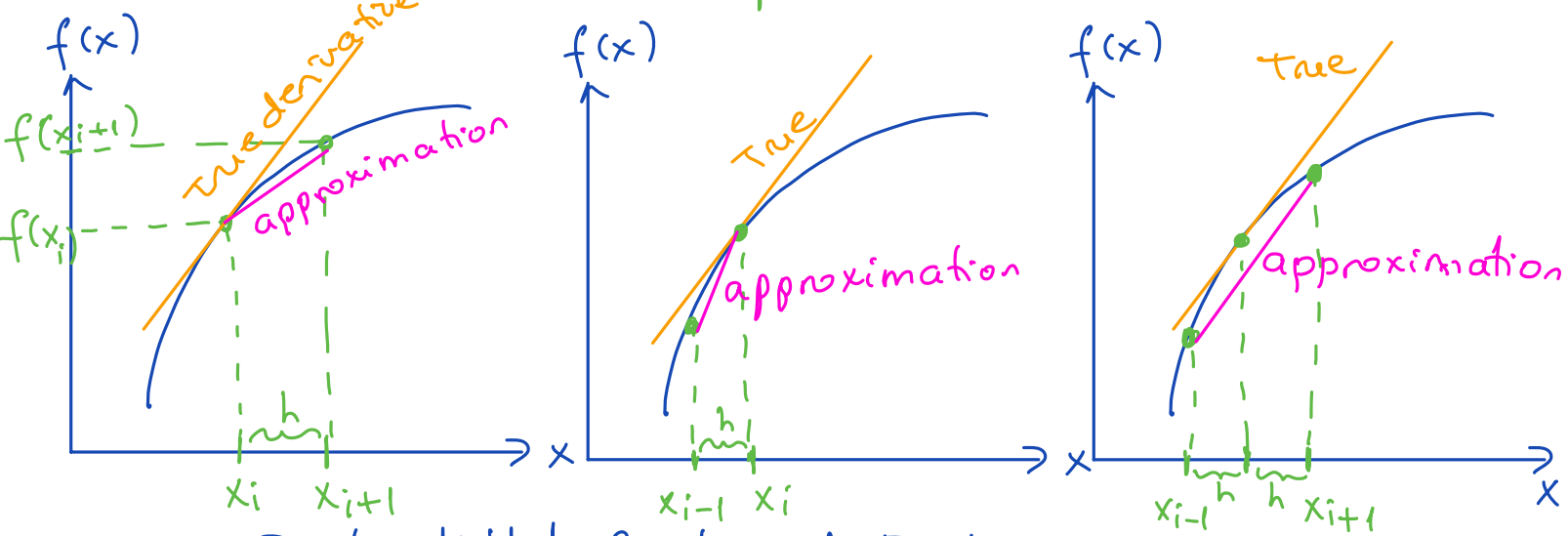
Forward Finite Divided Difference

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} + O(\overbrace{x_{i+1} - x_i}^{h})$$

$$f'(x_i) = \frac{\Delta f_i}{h} + O(h)$$

Δf_i → first forward difference

h → step size



Forward Finite-divided Difference

Backward Finite-Divided Diff

Centered finite divided difference

Backward Finite Divided Difference

$x_{i-1} - x_i = -h$

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \cdots \quad (1)$$

Backward Taylor Series

$$f'(x_i) \cong \frac{f(x_i) - f(x_{i-1})}{h} + O(h)$$

Error is $O(h)$

Centered Difference Approximation of the 1st order Derivative

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \dots \quad (2)$$

Subtract (1) from (2):

$$f(x_{i+1}) - f(x_{i-1}) = 2f'(x_i)h + \frac{2f^{(3)}(x_i)}{3!}h^3 + \dots$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} - O(h^2)$$

→ more accurate representation of the derivative

Ex Estimate the 1st derivative of (using forward, backward, centered methods)

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.25$$

at $x = 0.5$ using a step size of $h = 0.5$ and $h = 0.25$

True value: $f'(x) = -0.4x^3 - 0.45x^2 - x - 0.25$

$f'(0.5) = -0.925$

Approximations: $h = 0.5$ $\qquad x_{i+1} - x_i = h$, $x_i - x_{i-1} = h$

$x_{i-1} = 0$, $f(x_{i-1}) = 1.25$

$x_i = 0.5$, $f(x_i) = 0.925$

$x_{i+1} = 1$, $f(x_{i+1}) = 0.25$

Forward diff

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} = \frac{f(1) - f(0.5)}{1 - 0.5} = \frac{0.25 - 0.925}{0.5} = -1.45$$

$|\varepsilon_t| = 58.9\%$

Backward Diff

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} = \frac{f(0.5) - f(0)}{0.5 - 0} = \frac{0.925 - 1.25}{0.5} = -0.55$$

$|\varepsilon_t| = 39.7\%$

Centered Diff

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{x_{i+1} - x_{i-1}} = \frac{f(1) - f(0)}{1} = \frac{0.2 - 1.25}{1} = -1, \quad |\varepsilon_t| = 9.6\%$$

$h = 0.25$

$x_{i-1} = 0.25$, $f(x_{i-1}) = 1.103$

$x_i = 0.5$, $f(x_i) = 0.925$

$x_{i+1} = 0.75$, $f(x_{i+1}) = 0.6363$

Forward

$$f'(0.5) = \frac{f(0.75) - f(0.25)}{0.25} = -1.155 \Rightarrow |\varepsilon_t| = 26.5\%$$

(Decreased by a factor of 2)

Backward

$$f'(0.5) = \frac{f(0.5) - f(0.25)}{0.25} = -0.714 \Rightarrow |\varepsilon_t| = 21.7\%$$

Centered

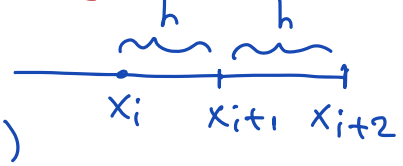
$$f'(0.5) = \frac{f(0.75) - f(0.25)}{0.5} = -0.934 \Rightarrow |\varepsilon_t| = 2.4\%$$

Decreased by a factor of 4.

more accurate than forward & backward.

Finite Order Approximations of Higher Derivatives

Forward Taylor series:



$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \cdots \quad (1)$$

Write a forward Taylor series expansion for $f(x_{i+2})$ i.t.o $f(x_i)$

$$f(x_{i+2}) = f(x_i) + f'(x_i)2h + \frac{f''(x_i)}{2!}(2h)^2 + \cdots \quad (2)$$

Multiply (1) by 2 and subtract from (2):

$$f(x_{i+2}) - 2f(x_{i+1}) = -f(x_i) + f''(x_i)h^2 + \cdots$$

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} + O(h)$$

Second forward Finite Difference

Second backward Finite Divided Difference

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2})}{h^2} + O(h)$$

Centered version (more accurate)

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2} + O(h^2)$$

$$= \frac{\dfrac{f(x_{i+1}) - f(x_i)}{h} - \dfrac{f(x_i) - f(x_{i-1})}{h}}{h}$$

Backward for 1st deriv.

Derivative of a derivative

Error Propagation

Goal: Study how errors in numbers can propagate through mathematical functions.

(e.g. if you multiply two numbers that have errors, estimate the error of the product)

Functions of a single variable

function : $f(x)$

$\tilde{x}$: an approximation of x

Goal: assess the effect of the discrepancy btwn x and $\tilde{x}$ on the value of the function

$$\Delta f(\tilde{x}) = |f(x) - f(\tilde{x})|$$ ↳ can't compute this:

$f(x)$ is unknown because x is unknown

overcome this: assume $\tilde{x}$ is close to x and $f(\tilde{x})$ is continuous and differentible. Use Taylor series around $\tilde{x}$:

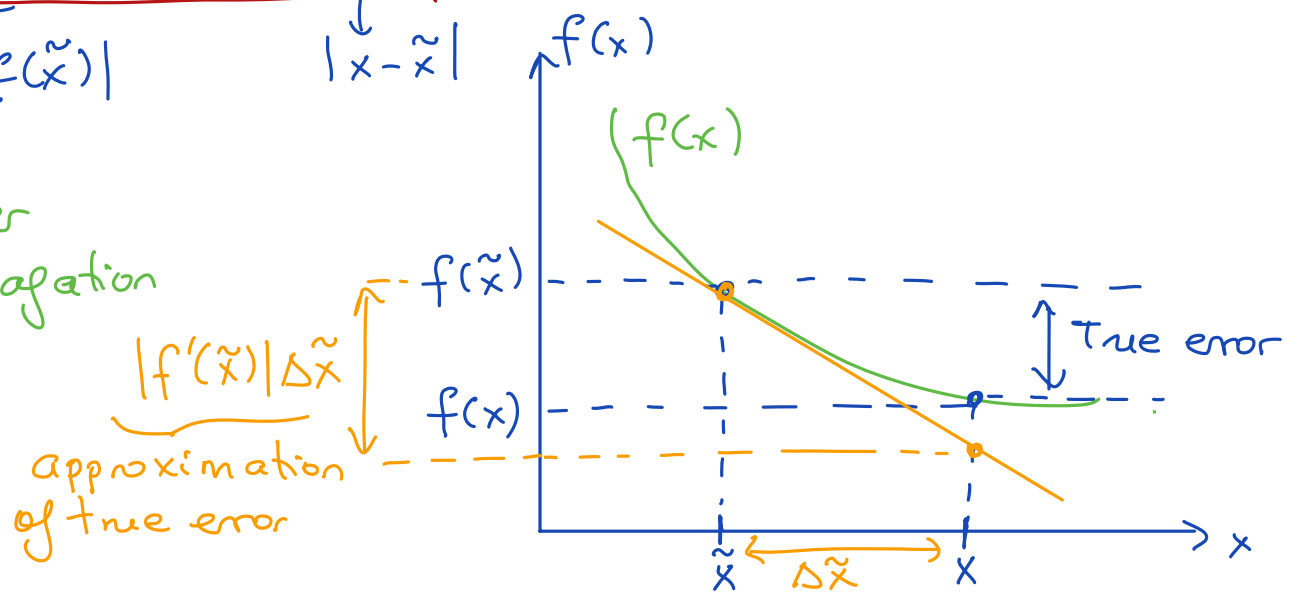
$$f(x) = f(\tilde{x}) + f'(\tilde{x})(x-\tilde{x}) + \underbrace{\frac{f''(\tilde{x})}{2!}(x-\tilde{x}) + \cdots}_{\text{Drop this}}$$

$$f(x) - f(\tilde{x}) = f'(\tilde{x})(x-\tilde{x})$$

$$\boxed{\Delta f(x) \cong |f'(\tilde{x})|\,\Delta\tilde{x}}$$

$\Delta f(x)$: $|f(x) - f(\tilde{x})|$; $\Delta\tilde{x}$: $|x - \tilde{x}|$

First order Error propagation



EX Given $\tilde{x} = 2.5$ with an error of $\Delta\tilde{x} = 0.01$, estimate the resulting error in the function $f(x) = x^3$.

$$\Delta f(\tilde{x}) \cong |f'(\tilde{x})| \Delta\tilde{x}$$

$$= 3\tilde{x}^2 \Big|_{\tilde{x}=2.5} \Delta\tilde{x} = 3(2.5)^2(0.01) = 0.1875$$

Since $f(2.5) = 15.625$ we can predict that

$$f(2.5) = 15.625 \pm 0.1875$$

or true value lies between 15.4375 and 15.8125

$$f(2.49) = 15.4375$$

$$f(2.51) = 15.8125$$

Functions of More Than One Variable

Function of two independent variables u and v, Taylor series can be written as:

$$f(u_{i+1}, v_{i+1}) = f(u_i, v_i) + \frac{\partial f}{\partial u}\Big|_{(u_i, v_i)} (u_{i+1} - u_i) + \frac{\partial f}{\partial v}\Big|_{(u_i, v_i)} (v_{i+1} - v_i)$$

$$+ \frac{1}{2!}\left[\frac{\partial^2 f}{\partial u^2}(u_{i+1} - u_i)^2 + 2\frac{\partial^2 f}{\partial u \partial v}(u_{i+1} - u_i)(v_{i+1} - v_i) + \frac{\partial^2 f}{\partial v^2}(v_{i+1} - v_i)^2 \right] + \cdots$$

all partial derivatives are evaluated at the base pt. i.

- Drop 2nd order terms and solve

$$\Delta f(\tilde{u}, \tilde{v}) = \left|\frac{\partial f}{\partial u}\right| \Delta\tilde{u} + \left|\frac{\partial f}{\partial v}\right| \Delta\tilde{v}$$

$\Delta\tilde{u}, \Delta\tilde{v}$: estimates of errors in u and v

Stability and Condition

- The condition of a mathematical problem relates to its sensitivity to changes in its input values.
- A computation is numerically unstable if the uncertainty in its values is grossly magnified by the numerical method.
  - Used a Taylor series

$$f(x) = f(\tilde{x}) + f'(\tilde{x})(x - \tilde{x})$$

Relative error of $f(x)$: $\dfrac{f(x) - f(\tilde{x})}{f(x)} \cong \dfrac{f'(\tilde{x})(x - \tilde{x})}{f(\tilde{x})}$

Relative error of x: $\dfrac{x - \tilde{x}}{\tilde{x}}$

$$\text{Condition number} = \frac{\text{Relative error of } f(x)}{\text{Relative error of } x} = \frac{\dfrac{f'(\tilde{x})(x - \tilde{x})}{f(\tilde{x})}}{\dfrac{x - \tilde{x}}{\tilde{x}}}$$

$$\text{Condition number (CN)} = \frac{\tilde{x}\, f'(\tilde{x})}{f(\tilde{x})}$$

- CN provides a measure of the extent to which the uncertainty in x is magnified by $f(x)$.

if $CN = 1 \Rightarrow$ The function relative error is identical to the relative error in x

if $CN > 1 \Rightarrow$ relative error is magnified

if $CN \gg 1 \Rightarrow$ the function is ill-conditioned

Ex Compute and interpret the condition number of

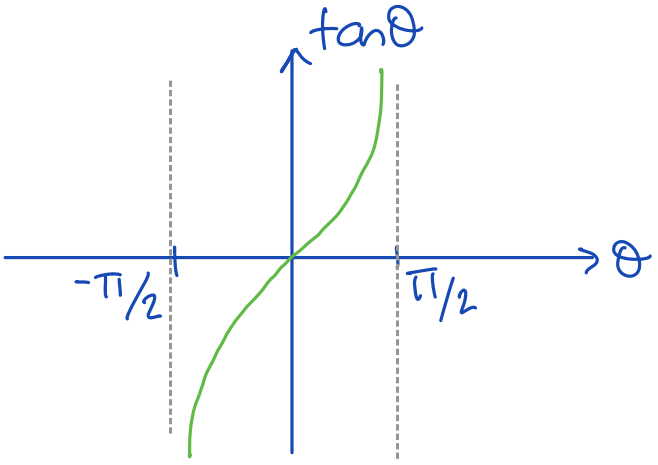
$f(x) = \tan x$ for $\tilde{x} = \frac{\pi}{2} + 0.1\left(\frac{\pi}{2}\right)$

$f(x) = \tan x$ for $\tilde{x} = \frac{\pi}{2} + 0.01\left(\frac{\pi}{2}\right)$

$$\text{Condition Number} = \frac{\tilde{x} f'(\tilde{x})}{f(\tilde{x})} = \frac{\tilde{x} \frac{1}{\cos^2 \tilde{x}}}{\tan \tilde{x}}$$

put $\tilde{x} = \frac{\pi}{2} + 0.1\left(\frac{\pi}{2}\right) \Rightarrow CN = -11.2 \Rightarrow$ Function is ill-conditioned

put $\tilde{x} = \frac{\pi}{2} + 0.01\left(\frac{\pi}{2}\right) \Rightarrow CN = -101 \Rightarrow$ "



- Major cause of ill-conditioning is the derivate at $\theta = \pi/2$